SESSION 6

# AGGREGATE FUNCTIONS

Solved SQL Tasks | Easy English Explanations

**Topics covered:** SUM(), COUNT(), AVG(), MIN(), MAX(), ROUND(), and GROUP BY.

## TASK 1 — SUM(): Total Amount Spent on Food Orders

Question: Calculate the total amount spent by users in the food_orders table (order_id, user_id, amount).

### SQL Query — Total for Each User

```
SELECT
    user_id,
    SUM(amount) AS TotalAmountSpent
FROM food_orders
GROUP BY user_id;
```

### Explanation

- SUM(amount) adds the amount values for the orders belonging to the same user.
- GROUP BY user_id creates one result row for each user.
- TotalAmountSpent is the name given to the calculated total.

This query shows how much each user has spent across their food orders.

### If you need one grand total for all users

```
SELECT SUM(amount) AS TotalAmountSpent
FROM food_orders;
```

This second version returns one value: the total amount across every row in food_orders.

## TASK 2 — COUNT(): Songs Added by Each User

Question: Count how many songs each user has added in spotify_playlists (playlist_id, user_id, song_id).

```
SELECT
    user_id,
    COUNT(song_id) AS SongsAdded
FROM spotify_playlists
GROUP BY user_id;
```

### Explanation

- COUNT(song_id) counts the non-NULL song_id values for each user.
- GROUP BY user_id groups the playlist records by user.
- SongsAdded is the output column name for the count.

The result contains one row per user represented in the table. If each row represents one song added, the count gives the number of songs recorded for that user.

## TASK 3 — AVG() and ROUND(): Average Movie Rating

Question: Calculate each movie's average rating in bookmyshow_reviews (review_id, movie_id, rating) and round it to one decimal place.

```
SELECT
    movie_id,
    ROUND(AVG(rating), 1) AS AverageRating
FROM bookmyshow_reviews
GROUP BY movie_id;
```

### Explanation

- AVG(rating) calculates the average rating for each movie.
- ROUND(AVG(rating), 1) rounds the average to one digit after the decimal point.
- GROUP BY movie_id calculates a separate average for each movie.

Illustrative example: if a movie's average rating is 4.267, the displayed AverageRating is 4.3.

## TASK 4 — MIN() and MAX(): Smallest and Largest Transaction

Question: Show the minimum and maximum transaction amount for each user in paytm_transactions (txn_id, user_id, amount).

```
SELECT
    user_id,
    MIN(amount) AS MinimumTransaction,
    MAX(amount) AS MaximumTransaction
FROM paytm_transactions
GROUP BY user_id;
```

### Explanation

- MIN(amount) returns the smallest transaction amount for each user.
- MAX(amount) returns the largest transaction amount for each user.
- GROUP BY user_id calculates these values separately for each user.

If you only want a specific user, add a WHERE condition such as WHERE user_id = 101 before GROUP BY.

## TASK 5 — Multiple Aggregate Functions with GROUP BY

Question: For each user in myntra_orders (order_id, user_id, total_price), show the number of orders, average order value rounded to two decimal places, and highest order value.

```
SELECT
    user_id,
    COUNT(order_id) AS TotalOrders,
    ROUND(AVG(total_price), 2) AS AverageOrderValue,
    MAX(total_price) AS HighestOrderValue
FROM myntra_orders
```

```
GROUP BY user_id;
```

### Explanation

- COUNT(order_id) counts the orders for each user.
- AVG(total_price) calculates each user's average order value.
- ROUND(AVG(total_price), 2) rounds the average to two decimal places.
- MAX(total_price) returns that user's highest order value.
- GROUP BY user_id produces one result row per user.

## Final Revision Table

| Function | Purpose |
|---|---|
| **SUM()** | Adds values together. |
| **COUNT()** | Counts rows or non-NULL values. |
| **AVG()** | Calculates the average. |
| **ROUND(value, decimals)** | Rounds a number to the specified decimal places. |
| **MIN()** | Returns the smallest value. |
| **MAX()** | Returns the largest value. |
| **GROUP BY** | Calculates aggregate results separately for each group. |

**Key idea:** Aggregate functions summarize data. Use GROUP BY when you need the summary separately for each user, movie, or other category.